

INFO 5507

Data and the Humanities

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Machine Learning Engineer (49.3 / 100)	41.0 / 100

41	86	+55
CURRENT AVG SCORE	POTENTIAL SCORE	MAX SCORE GAIN
across 6 job roles	with recommended additions	best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces students to foundational computing and statistical concepts for analyzing humanities data. This course discusses the influence of digitization and data on humanist inquiry and exposes students to techniques for working with data in different areas of the humanities, including literature, history, and art. The course emphasizes technical practices involved in humanist data analysis. Comfort with programming is strongly encouraged.
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INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	49.3	85.1	+35.8	HIGH
AI Engineer	46.5	94.8	+48.3	HIGH
Data Scientist	44.4	94.3	+49.8	HIGH
ML Engineer	43.0	75.1	+32.1	HIGH
Applied Scientist	38.9	93.9	+55.0	HIGH
MLOps Engineer	24.0	74.0	+50.0	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

Deep Learning

1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

2 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Machine Learning Engineer

49.3 → 85.1 (+35.8 pts)

sets learning tools exploratory design development science models

AI Engineer

46.5 → 94.8 (+48.3 pts)

perception robotics models learning build software model systems

Data Scientist

44.4 → 94.3 (+49.8 pts)

toolkit sets mitigation social build

ML Engineer

43.0 → 75.1 (+32.1 pts)

learning machine systems models model build new building

Applied Scientist

38.9 → 93.9 (+55.0 pts)

biology computational research models learning systems industry design

MLOps Engineer

24.0 → 74.0 (+50.0 pts)

detection model text systems learning machine image training

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.